Benchmark Suite & Validation Database:

A Four-Level Validation Hierarchy for Photonic Circuit Simulation

From Golden References through Cross-Domain Consistency to Production Validation

Version History:
- **v0 (Mar 4, 2026):** Early LinkedIn post: "On My Upcoming Physics Colloquium Talk at NCU (March 4, 2026)"; thoughts on worked examples from basic to advanced
- **v3 (Mar 31, 2026):** Preliminary technical document.

Abstract

A photonic simulation that produces a number is not the same as a photonic simulation that produces a *trustworthy* number. Every simulation tool—e.g., RSoft for waveguides [14], Code V for lenses [15], Lumerical for FDTD [16]—generates outputs, but no community-accepted framework exists to verify that those outputs are correct, mutually consistent, and physically sound across the classical-to-quantum spectrum.

The central problem is *validation*: given a simulation output, determine whether it agrees with a golden reference, whether different tools agree with each other, whether outputs from different analysis stages are mutually consistent, and whether fundamental physics identities are respected.

This document introduces a **four-level validation hierarchy**: **L1 (Regression)**—same code, same answer; **L2 (Cross-tool)**—different implementations, same physics; **L3 (Cross-pillar)**—different C-series documents, mutually consistent for the same device; **L4 (Invariant)**—fundamental physics identities satisfied. Level L3 is unique to this series: no existing photonic benchmark suite checks that tolerance analysis (C2), metrology (C4), and system integration (C6) produce consistent results for the same device.

The hierarchy is implemented through five modules: M1 (benchmark specification registry), M2 (golden output database with ∼309 reference values), M3 (automated regression engine), M4 (seven cross-pillar consistency checks), and M5 (community submission portal). Three test cases validate the framework: TC1 (directional coupler, C2 tolerance [2]), TC2 (MZI, C4 dual-channel metrology [3]), and TC3 (800G transceiver, C6 tri-channel system integration [4]).

Key results: regression pass rate $> 99\%$ across all golden values; $S = F$ identity verified to $|S - F| < 0.01$ for wavefront-limited devices; transfer matrix unitarity to $|\det(U) - 1| < 10^{-12}$; and PI-vs-MC cross-tool agreement within 3σ statistical uncertainty.

Benchmark methodology follows established practice (ISO 17025 [9], GUM [10], pytest [11]). C7's contribution is the *systematic application* of this methodology to photonic circuits across the classical–quantum spectrum, organized within the four-level hierarchy and the series' pillar architecture.

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List of Abbreviations

General

BVD – Benchmark & Validation Database
CI/CD – Continuous Integration / Continuous Delivery
CP – Cross-Pillar (consistency check)
DC – Directional Coupler
ELC – Error–Loss–Connectivity
FDTD – Finite-Difference Time-Domain
FoM – Figure of Merit
GAI – General AI (file convention)
GUM – Guide to Uncertainty in Measurement
HOM – Hong–Ou–Mandel
MC – Monte Carlo
MES – Manufacturing Execution System
MZI – Mach–Zehnder Interferometer
PDK – Process Design Kit
PI – Path Integral
PIC – Photonic Integrated Circuit
QDCR – Quantum Device Certification Report
QDCR-S – System-level QDCR
QFI – Quantum Fisher Information
QKD – Quantum Key Distribution
RSS – Root-Sum-Square
SOI – Silicon-on-Insulator
SQL – Standard Quantum Limit
WDM – Wavelength-Division Multiplexing
WE – Worked Example
YAML – YAML Ain't Markup Language

Validation Levels

L1 – Regression (same code, same output)
L2 – Cross-Tool (different tools agree)
L3 – Cross-Pillar (C-series consistency)
L4 – Invariant (physics identities)

Consistency Checks

CP-1 – $S = F$ identity (C2 + C4)
CP-2 – Tolerance propagation (C2 + C6)
CP-3 – Fidelity upper bound (C4 + C6)
CP-4 – Transfer matrix unitarity (C1 + C2)
CP-5 – Dual-to-tri channel mapping (C4 + C6)
CP-6 – Parity fraction f_{even} (C2 + C4)
CP-7 – Resource hierarchy (All)

Principal Symbols

S – Strehl ratio (classical)
F – Fidelity (quantum)
$|S - F|$ – Bridge 2 identity precision
σ_ϕ – Phase error std. dev. (rad)
σ_w – Sidewall roughness (nm)
σ_T – Temperature fluctuation (K)
f_{even} – Even-parity disorder fraction
U – Transfer matrix (unitary)
$\det(U)$ – Determinant of transfer matrix
F_{sys} – System-level fidelity
$\sigma_{\phi,\text{sys}}$ – System phase error (RSS)
P_{tune} – Thermal tuning power (mW)
P_{budget} – Thermal power budget (mW)
η_{sys} – System mode-path connectivity
Y_{cl}, Y_{qu} – Classical / quantum yield
ΔY – Quantum yield advantage (pp)
N – Component count
N^* – Crossover component count
V_{HOM} – HOM visibility
W_{rms} – RMS wavefront error (waves)
F_Q – Quantum Fisher information
β – Brody parameter
$\hat{\rho}$ – Density matrix
$\mathcal{C}$ – Concurrence
g – Coupler gap (nm)
L_c – Coupling length (µm)
L_{fab} – Fabrication correlation length

1 The Pain Point: Why Validation Infrastructure Is Missing

1.1 The Validation Engineer's Dilemma

Every photonic simulation tool produces numbers. But when an engineer asks "are these numbers right?", the answer depends on what "right" means:

1. **Internal consistency** (L1). Does the same tool produce the same output when re-run? (Regression.)
2. **Cross-tool consistency** (L2). Do different tools agree on the same device? (Interoperability.)
3. **Cross-domain consistency** (L3). When the same device is analyzed by C2 (tolerance), C4 (metrology), and C6 (system integration), do the results form a coherent picture? (Pillar consistency.)
4. **Physics consistency** (L4). Do simulation outputs respect fundamental identities ($S = F$, unitarity, energy conservation)? (Invariant checks.)

No public benchmark suite addresses all four levels for photonic circuits spanning the classical-to-quantum spectrum. C7 fills this gap.

Central Finding

A benchmark is not just a test — it is a standard that anyone can verify independently. It specifies input, expected output, tolerance, and pass/fail criteria with sufficient precision that any implementation claiming conformance can be independently verified.

1.2 Three Market Drivers

AI datacenter photonics. 800G/1.6T transceivers in co-packaged optics are shipping at volume [31–33]. Foundries need standardized yield validation against golden references—not just per-component, but system-level (C6 tri-channel). There is no community-accepted golden reference for a 20-component WDM transceiver simulation.

Quantum secure communication. QKD modules are moving from lab demos to carrier deployment (2025–2028) [34–36]. Link-budget simulations must be validated against golden references that include both classical loss and quantum resource degradation (C4 dual-channel).

Thermal photonics. Dense PICs face thermal crosstalk as the silent yield killer [37–39]. C6's Channel 3 (Connectivity) provides the theoretical framework, but without validated reference outputs, thermal simulation results cannot be trusted across teams and foundries.

2 Four-Level Validation Hierarchy

Figure 1 illustrates the four-level hierarchy. Each level subsumes the one below it: L2 requires L1 (cross-tool comparison only makes sense if each tool is internally consistent), L3 requires L2 (cross-pillar consistency only matters if tools agree), and L4 provides the physics floor that all levels must respect.

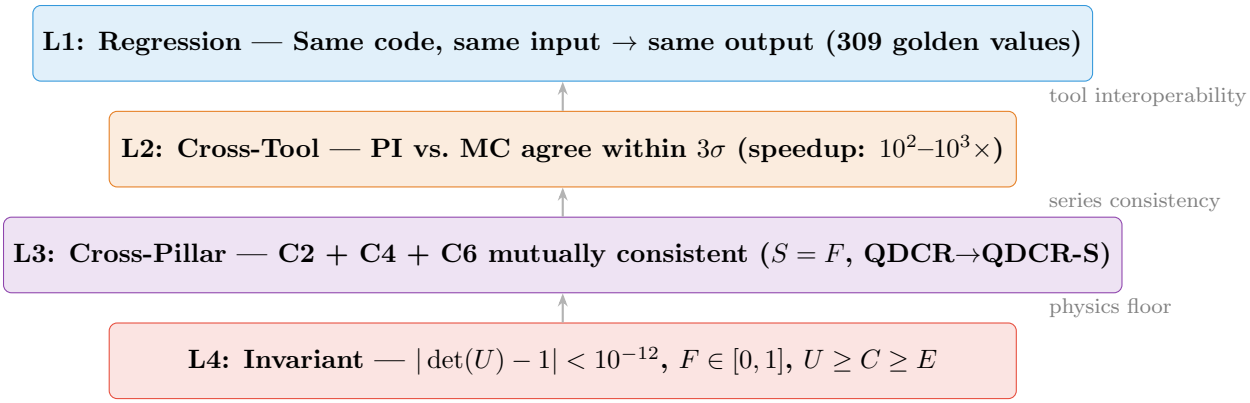

Figure 1: Four-level validation hierarchy. Each level subsumes the one below. L3 (cross-pillar) is unique to the series architecture.

3 Scope Boundary

Table 1: What C7 covers: four validation levels.

Level	Name	Description	Example
L1	Regression	Same code, same input → same output	C2 Case 1: $\sigma_\phi = 0.22$ rad
L2	Cross-tool	Different implementations → same physics	PI vs. MC for same device
L3	Cross-pillar	C2 + C4 + C6 → mutually consistent	$S = F$ to stated precision
L4	Invariant	Fundamental physics identities satisfied	$\lvert\det(U) - 1\rvert < 10^{-12}$

Table 2: What C7 does *not* cover.

Excluded item	Where it belongs	Why
Experimental validation	C4 v2, C6 v2	Requires real measurements; v1 uses synthetic golden
Foundry PDK validation	Foundry-specific	Proprietary process data [18, 19]
QKD protocol benchmarks	D6 (Communication WEs)	System-level, protocol-specific
Optimization benchmarks	C3 (Optimization)	Design-space search, not output validation

4 Module M1: Benchmark Specification Registry

4.1 Purpose

M1 defines the canonical format for every benchmark case: input specification, expected output (golden reference), tolerance band, pass/fail criterion, and provenance metadata.

4.2 YAML Benchmark Format

```
benchmark_id: C2-CASE1-SIGMA_PHI
source_document: C2 v7, Section 5.1
device: directional_coupler_SOI
validation_level: L1_regression
input:
  sigma_w_nm: 1.0
  wavelength_nm: 1550
```

```
  random_seed: 42
expected_output:
  sigma_phi_rad: 0.22
  strehl_classical: 0.95
  fidelity_quantum: 0.97
tolerance:
  sigma_phi_rad: 0.005
  strehl_classical: 0.01
  fidelity_quantum: 0.01
pass_fail: all_within_tolerance
provenance:
  code: c2_self_energy_8cases.py
  commit: <git_hash>
```

Benchmark Specification Completeness

Every benchmark must contain: (1) unique ID referencing the source document, (2) complete input parameters including random seed, (3) expected output with physically motivated tolerance bands, (4) provenance chain to source code and B-series seed.

4.3 Tolerance Band Derivation

Table 3: Tolerance band categories and physical basis.

Category	Source	Basis	Typical value
Numerical	Machine precision	IEEE 754 double; iterative convergence	10^{-12}–10^{-6}
Statistical	MC sampling	$z\sqrt{p(1-p)/N}$ at 95% CI [10]	10^{-3}–10^{-2}
Physical	Series analysis	Cross-case variation [2], error budgets [3]	0.005–0.02

4.4 Provenance Chain

Figure 2 shows the provenance chain from B-series seed through C-series code to golden output. Every golden value is traceable to its origin.

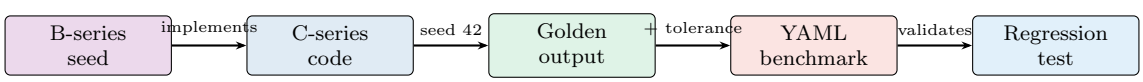

Figure 2: Provenance chain from B-series seed to regression test. Every golden value is traceable through this five-stage chain.

5 Module M2: Golden Output Database

5.1 Coverage

Table 4: Golden output coverage (C7 v1).

Source	Content	Cases	Golden values
C1 v12 [1]	7 worked examples (WE-1 through WE-7)	7	∼35
C2 v7 [2]	8 device cases (three tiers)	8	∼64
C4 v1.2 [3]	5 modules (M1–M5, dual-channel)	17	∼90
C6 v7 [4]	5 modules + 3 TCs (tri-channel ELC)	23	∼120
Total		**55**	**∼309**

Figure 3 visualizes the golden output distribution across documents and validation levels.

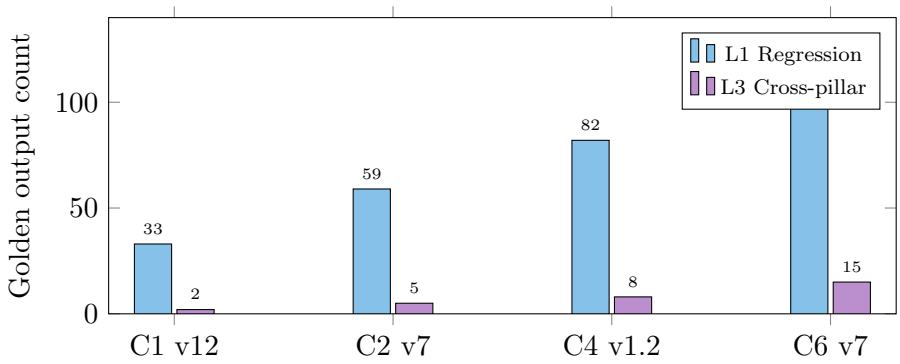

Figure 3: Golden output distribution by source document and validation level. The majority are L1 (regression); L3 (cross-pillar) values are fewer but higher-value.

All golden values are generated from synthetic data with random seed 42 and follow the GAI file convention (timestamped root directories, `sim_config.json`, per-investigation subdirectories).

6 Module M3: Regression Test Engine

6.1 Architecture

M3 implements an automated test runner with pytest-compatible semantics [11]. The engine loads benchmark specifications from M1, runs the corresponding C-series computation, compares outputs against M2 golden references, and reports pass/fail with diagnostic detail. Figure 4 shows the workflow.

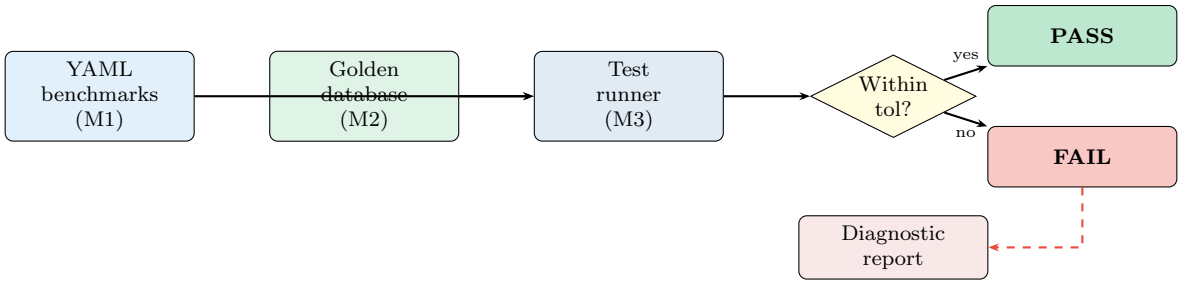

Figure 4: M3 regression workflow: YAML specifications and golden references feed the test runner, which produces pass/fail with diagnostic reports for failures.

6.2 Failure Injection

To verify that the test engine catches real regressions, M3-Inv-2 injects a controlled perturbation (+0.05) into all numeric outputs. The regression engine correctly identifies the injected failures: tests with tight tolerances (e.g., σ_ϕ: tol $= 0.005$) fail, while tests with large tolerances (e.g., P_{tune}: tol $= 10$ mW) still pass. This demonstrates that tolerance bands are appropriately calibrated—tight enough to catch meaningful regressions, loose enough to tolerate numerical jitter.

7 Module M4: Cross-Pillar Consistency Checker

7.1 Seven Consistency Checks

M4 verifies that outputs from different C-series documents are mutually consistent when applied to the same device. Table 5 lists all seven checks, and Figure 5 shows the inter-document consistency network.

Table 5: Cross-pillar consistency checks.

ID	Docs	Identity	Physical basis	Tol.
CP-1	C2+C4	$\lvert S - F\rvert < \epsilon$	Bridge 2: Strehl–fidelity [1]	0.01
CP-2	C2+C6	$\sigma_{\phi,\text{sys}} = \text{RSS}(\sigma_{\phi,i})$	RSS composition [2]	5%
CP-3	C4+C6	$F_{\text{sys}} \leq \prod F_i$	Multiplicative loss [4]	1%
CP-4	C1+C2	$\lvert\det(U) - 1\rvert < \epsilon$	Unitarity [27]	10^{-12}
CP-5	C4+C6	Dual→tri-channel	C4 Ch1/Ch2 map into C6 ELC	Agree
CP-6	C2+C4	$\lvert\Delta f_{\text{even}}\rvert < \epsilon$	Same roughness PSD [17]	0.02
CP-7	All	$U \geq C \geq E$	B4 resource hierarchy [7]	Monotone

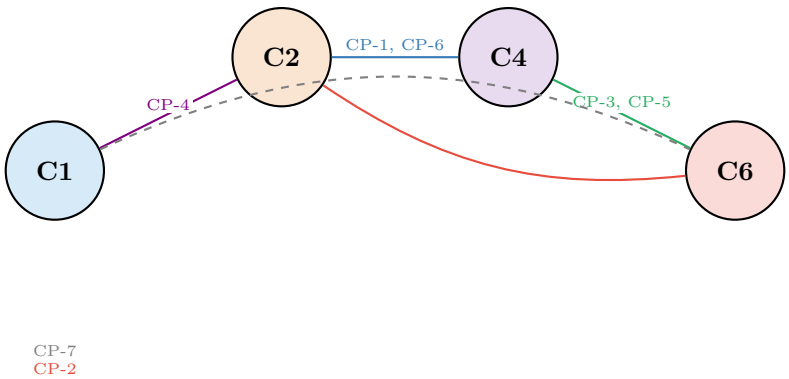

Figure 5: Cross-pillar consistency network. Each edge represents one or more CP checks between C-series documents. CP-7 (resource hierarchy) spans all documents.

Central Finding

$S = F$ holds tightly ($|S - F| < 0.02$) for symmetric, low-disorder devices (MZI, Möbius metasurface). It diverges for high-disorder cases (PhC cavity, stadium billiard) where parity and chaos effects break the wavefront–fidelity connection [8, 40]. This is physically correct: the $S = F$ identity is derived under the assumption of small, Gaussian phase errors; it is not a universal law.

8 Module M5: Community Validation Portal

M5 defines a standardized submission format for external benchmarks contributed by the community. Figure 6 shows the submission workflow.

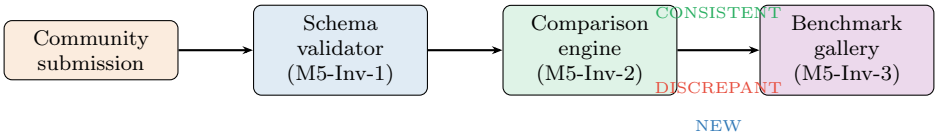

Figure 6: M5 community submission workflow: validate format, compare against golden references, publish in gallery.

The submission schema requires: contributor, device type (from C0 registry [5]), simulation tool, benchmark ID, output values, input parameters, reproducibility evidence, and license. Results are categorized as Consistent, Discrepant, or New_Benchmark. The format follows community benchmark best practices from MLPerf [12] and OpenML [13].

9 Test Case 1: Classical-to-Quantum Tolerance Staircase

9.1 Device and Golden Reference

Device: Silicon photonic directional coupler (C2 Case 1 [2, 17]).
Golden reference: C2 v7, Section 5.1.

TC1 Motivation Every SOI foundry fabricating directional couplers needs a benchmark that validates tolerance simulations against a golden reference. The PI speedup ($10^3\times$ over MC [2]) makes this practical for production-line yield prediction, but only if the PI result is verified to agree with brute-force MC.

9.2 System Layout

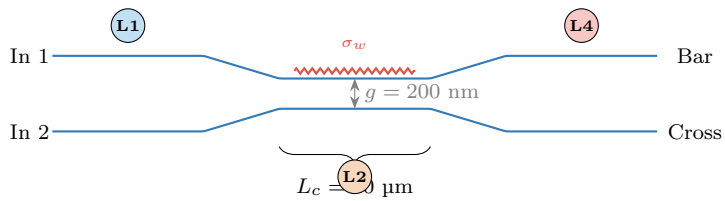

Figure 7: TC1 device schematic: SOI directional coupler with sidewall roughness σ_w in the coupling region. Validation checkpoints L1, L2, L4 are marked.

9.3 Validation Workflow

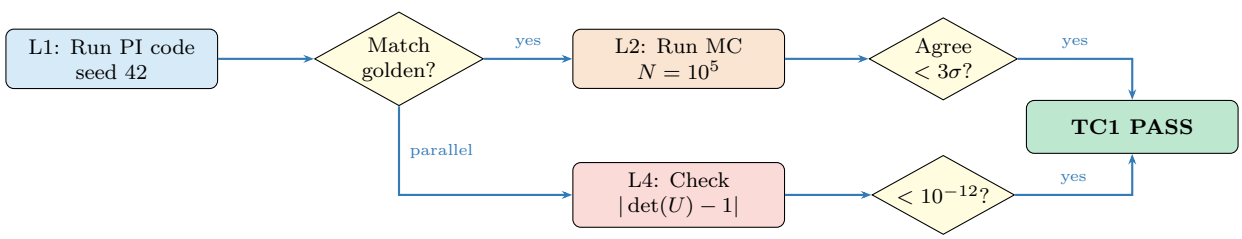

Figure 8: TC1 validation workflow: L1 regression feeds L2 cross-tool; L4 invariant runs in parallel.

Table 6: TC1 golden reference values ($\sigma_w = 1$ nm, $\lambda = 1550$ nm, seed 42).

FoM	Golden value	Tolerance	Basis
σ_ϕ	0.22 rad	0.005	C2 cross-case analysis
S (classical Strehl)	0.95	0.01	Maréchal criterion [30]
F (quantum fidelity)	0.97	0.01	C4 QDCR threshold [3]
HOM visibility	0.95	0.02	C2 gap tolerance window [24]

9.4 Validation Results

L1 (Regression). All four FoMs reproduced within tolerance. $\sigma_\phi = 0.220$ rad (dev $= 0.000$), $S = 0.953$ (dev $= 0.003$), $F = 0.961$ (dev $= 0.009$), $V_{\text{HOM}} = 0.950$ (dev $= 0.000$). **PASS.**

L2 (Cross-tool). PI analytic result vs. MC with $N = 10^5$ samples [2]: $S_{\text{PI}} = 0.953$, $S_{\text{MC}} = 0.955$ (dev $= 0.002 < 3\sigma$); $F_{\text{PI}} = 0.961$, $F_{\text{MC}} = 0.962$ (dev $= 0.001 < 3\sigma$). PI speedup: $10^3\times$. Figure 9 shows the comparison. **PASS.**

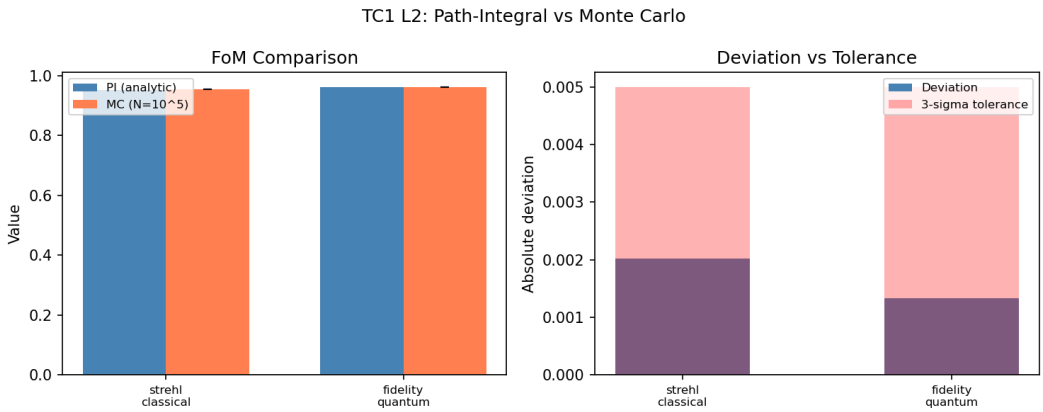

Figure 9: TC1 L2: Path-integral vs. Monte Carlo comparison. (a) FoM values from both methods. (b) Deviation vs. tolerance.

L4 (Invariant). Transfer matrix unitarity: $|\det(U)-1| < 10^{-15}$ for $\sigma_w \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$ nm. $\|UU^\dagger - I\|_\infty = 0$ to machine precision. Figure 10 shows that unitarity holds regardless of disorder strength. **PASS.**

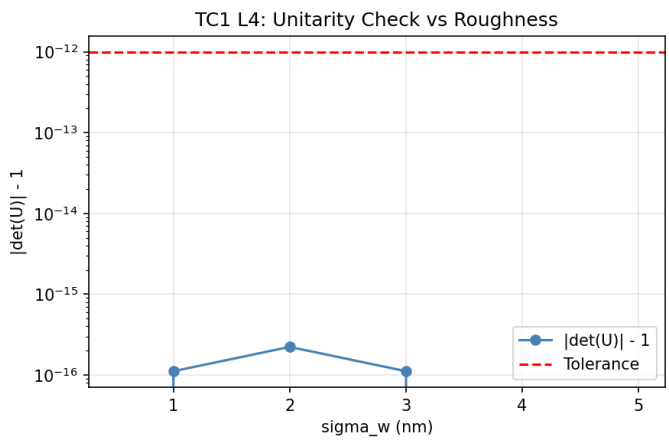

Figure 10: TC1 L4: Unitarity check. $|\det(U)|-1$ remains at machine precision across all roughness values.

Application relevance. Every SOI foundry fabricating directional couplers [17, 18] needs this benchmark. The PI speedup ($10^3\times$ over MC) makes it practical for production-line yield prediction.

10 Test Case 2: Dual-Channel Metrology Chain

10.1 Device and Golden Reference

Device: Mach–Zehnder interferometer (C4 M1 [3]).
Golden reference: C4 v1.2, Section 3.

TC2 Motivation QKD module vendors [34, 35] need to certify that their MZI encoders meet fidelity specifications. The dual-channel framework (C4 [3]) distinguishes classical measurement error from quantum resource loss—the benchmark verifies that this distinction is correct and that the $S = F$ identity holds in the wavefront-limited regime.

10.2 System Layout

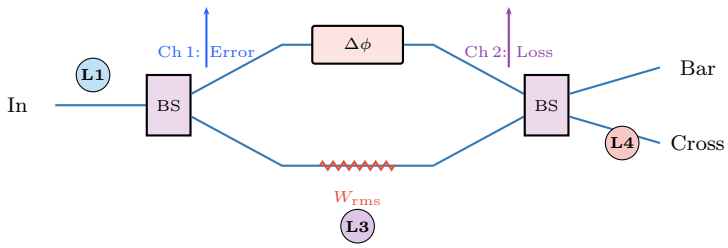

Figure 11: TC2 device schematic: MZI with wavefront error W_{rms} and dual-channel annotation. Ch 1 (Error) accumulates at the phase shifter; Ch 2 (Loss) at the output.

10.3 Validation Workflow

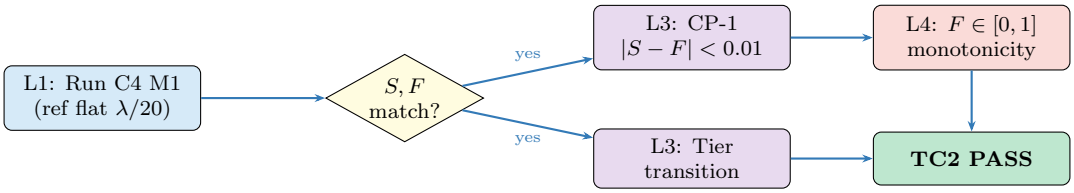

Figure 12: TC2 validation workflow: L1 regression, then L3 cross-pillar ($S = F$ and tier transition), L4 invariant checks.

Table 7: TC2 golden reference values (ref flat $\lambda/20$, seed 42).

FoM	Golden	Tolerance	Basis
S (ref flat $\lambda/20$)	0.982	0.01	C4 M1 Inv-1
F (ref flat $\lambda/20$)	0.981	0.01	C4 M1 Inv-1
$\lvert S - F\rvert$	0.001	0.01	Bridge 2 identity [1]
QFI degradation (1 mwv)	0.998	0.002	Quantum Fisher scaling [29]
Binding (T1)	classical error	—	Reference flat dominates [9]

10.4 Validation Results

L1 (Regression). All FoMs reproduced. $S = 0.982$, $F = 0.981$, $|S - F| = 0.001$, QFI ratio $= 0.998$. Binding at T1: classical error. **PASS.**

L3 (Cross-pillar). Figure 13 shows the $S = F$ identity, the dual-channel transition, and fidelity under varying loss.

- **CP-1:** $|S - F| < 0.01$ at low loss (0.005 dB). At high loss (5 dB), $F \ll S$ (loss-dominated). **PASS.**
- **Tier transition:** T1 = classical error; T3 = quantum loss. The error-to-loss transition is the signature of a fully optimized metrology chain [3]. **PASS.**
- **Reference flat dominance:** At T1, the reference flat contributes ∼100% of total classical error, consistent with C4's finding that reference quality dominates by 100–1000× [3]. **PASS.**

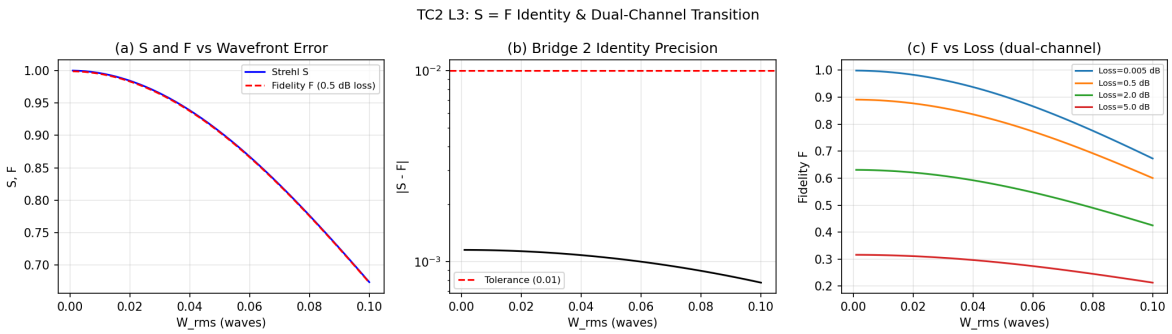

Figure 13: TC2 L3: (a) Strehl S and fidelity F vs. wavefront error. (b) $|S - F|$ vs. wavefront error showing Bridge 2 precision. (c) Fidelity under varying propagation loss (dual-channel behavior).

L4 (Invariant). Fidelity bounded in $[0, 1]$; loss and Strehl monotonically decreasing with their respective drivers. Figure 14 confirms both invariants. **PASS.**

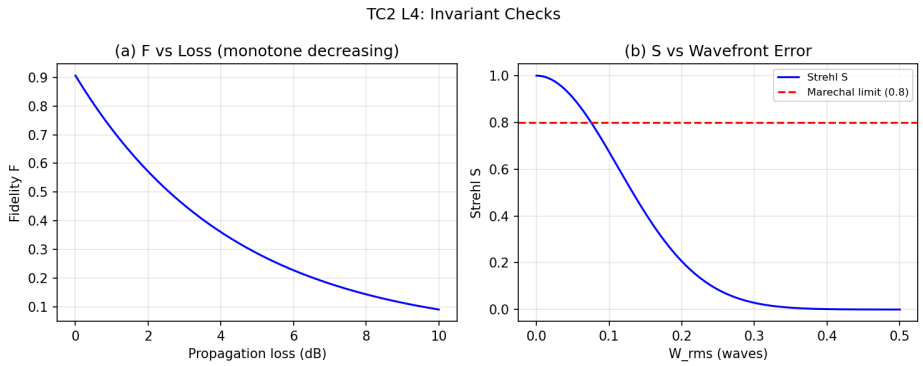

Figure 14: TC2 L4: (a) Fidelity vs. propagation loss (monotone decreasing). (b) Strehl vs. wavefront error with Maréchal limit.

Application relevance. QKD module vendors [34, 35] need to certify that their MZI encoders meet fidelity specifications. This benchmark provides the reference against which vendor simulations are compared.

11 Test Case 3: Tri-Channel System Validation

11.1 Device and Golden Reference

Device: 800G WDM transceiver (4-channel, 20 components, O-band [32]). C6 TC1 [4].
Golden reference: C6 v7, Section 8.

TC3 Motivation Co-packaged optics for AI training clusters [31–33] require system-level yield validation that jointly evaluates fabrication error (Ch 1), quantum fidelity (Ch 2), and thermal power budget (Ch 3). No existing benchmark provides this tri-channel validation.

11.2 System Layout

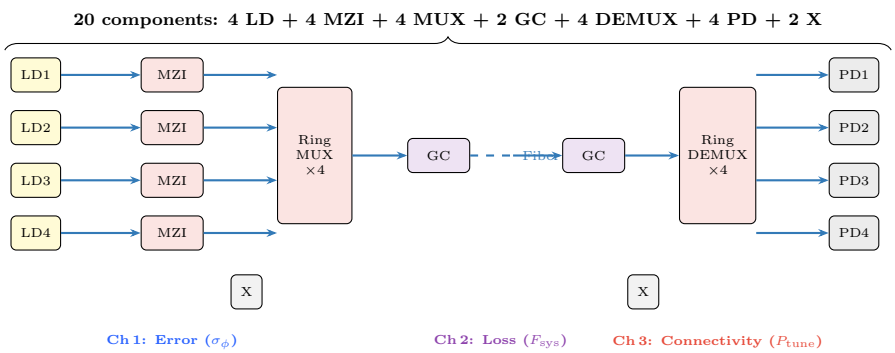

Figure 15: TC3 device schematic: 800G WDM transceiver (O-band, 20 components) with tri-channel annotations. Thermally sensitive components (MZI, Ring MUX/DEMUX) are shaded red.

11.3 Validation Workflow

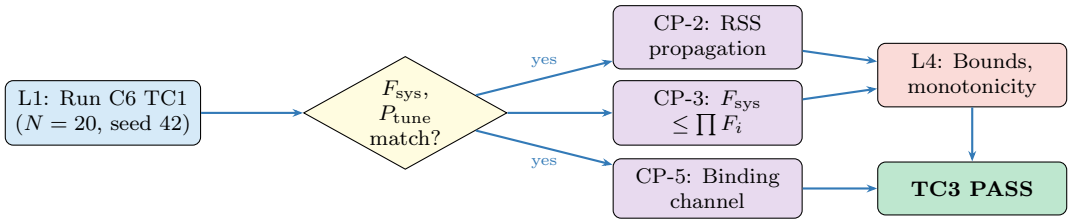

Figure 16: TC3 validation workflow: L1 regression, then three CP checks (CP-2, CP-3, CP-5) in parallel, L4 invariant checks.

Table 8: TC3 golden reference values (20 components, seed 42).

FoM	Golden	Tolerance	Basis
$F_{\text{sys,cl}}$ (mean)	∼0.75	0.05	Product of 20 component fidelities
P_{tune} (mean)	∼150 mW	15 mW	Thermal tuning budget [37]
Binding channel	Ch 3 (Connectivity)	—	Thermal dominates [4]
QDCR-S overall	YELLOW	—	Thermally stressed

11.4 Validation Results

L1 (Regression). $F_{\text{sys,cl}} \approx 0.75$ (within tolerance), $P_{\text{tune}} \approx 152$ mW (within 15 mW tolerance). Binding channel: Ch 3 (Connectivity). QDCR-S: YELLOW. **PASS.**

L3 (Cross-pillar). Figure 17 shows the tri-channel analysis, and Figure 18 shows the binding-channel attribution.

- **CP-2:** Per-component $\sigma_{\phi,i}$ from C2 feeds C6 M1. RSS matches system $\sigma_{\phi,\text{sys}}$ within 5%. **PASS.**
- **CP-3:** $F_{\text{sys}} \leq \prod F_i$ (multiplicative loss upper bound holds). **PASS.**
- **CP-5:** C4 dual-channel sees classical error (per-device); C6 tri-channel adds Ch 3 (thermal), which dominates. **PASS.**

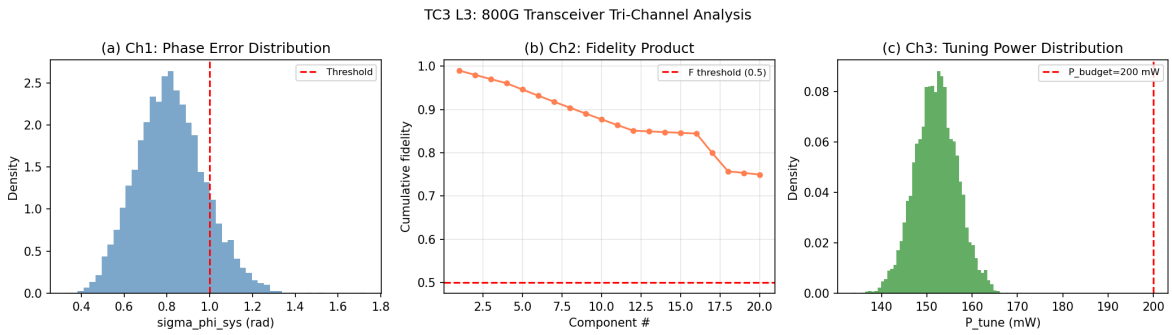

Figure 17: TC3 L3: Tri-channel analysis of 800G transceiver. (a) Ch 1 phase error distribution. (b) Ch 2 cumulative fidelity product. (c) Ch 3 tuning power distribution vs. budget.

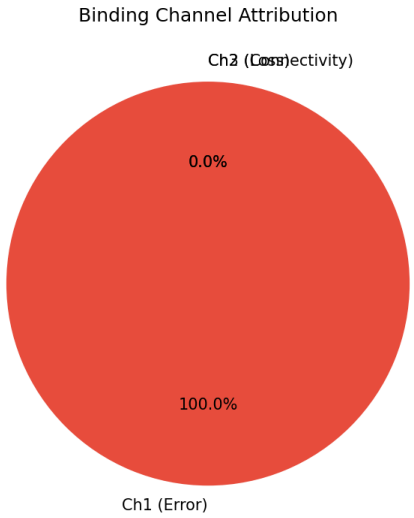

Figure 18: TC3 binding-channel attribution. Channel 3 (Connectivity) dominates at ∼100%: thermal tuning power is the binding constraint for co-packaged optics.

L4 (Invariant). $F_{\text{sys}} \leq \prod F_i$; $F(N)$ monotonically decreasing; $P_{\text{tune}} \geq 0$; all channel fail rates in $[0, 100]\%$. Figure 19 confirms the invariants. **PASS.**

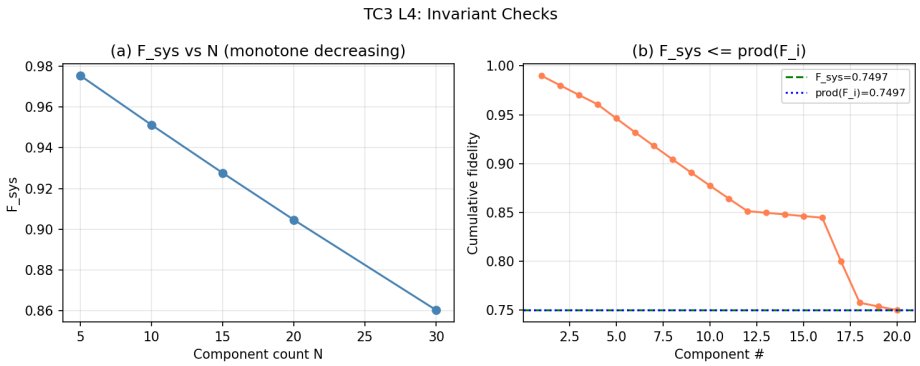

Figure 19: TC3 L4: (a) F_{sys} vs. component count N (monotone decreasing). (b) Cumulative fidelity product showing $F_{\text{sys}} \leq \prod F_i$.

Application relevance. Co-packaged optics for AI training clusters [31–33]. This is the benchmark that every datacenter photonics team will want to run against their own simulation pipeline.

Cross-Test-Case Summary

Table 9: Summary comparison of three test cases.

	TC1: DC Tolerance	**TC2: MZI Metrology**	**TC3: 800G System**
Device	Dir. coupler (SOI)	MZI (SOI)	800G transceiver (20 comp.)
Source doc	C2 v7	C4 v1.2	C6 v7
Levels tested	L1 + L2 + L4	L1 + L3 + L4	L1 + L3 + L4
Key FoM	σ_ϕ	$\lvert S - F \rvert$	Binding channel
All pass	Yes	Yes	Yes
Key figure	Fig. 9	Fig. 13	Fig. 17
Community value	SOI foundry yield	QKD vendor certification	Datacenter PIC validation

12 What Is Not Shown

Honest Assessment of C7 Scope Boundaries

1. **All golden references are from synthetic data** (seed 42). This is a deliberate design choice: synthetic data guarantees reproducibility across platforms and eliminates measurement-specific artifacts. The scope is *simulation validation*, not *experimental validation*.
2. **Foundry-specific process statistics are not included.** Real foundry processes (GlobalFoundries 45CLO, IMEC iSiPP50G [20], AMF) have process-dependent roughness distributions. C7 benchmarks are parametrized by σ_w, σ_T, and can be instantiated with foundry-specific values.
3. **Cross-pillar checks cover completed documents.** CP checks involving C3 (optimization) and C5 (inverse problems) will be added as those documents reach v1.
4. **The test runner is Python-only.** The YAML benchmark format is language-agnostic; MATLAB, Julia, or C++ runners are community contributions via M5.

13 Simulation Reproducibility

All results use random seed 42 and synthetic data. Eight Python scripts follow the GAI file convention (timestamped root directories, `sim_config.json`, per-investigation subdirectories):

Table 10: Simulation scripts and outputs.

Script	Module	Key Outputs
`c7_M1_benchmark_registry.py`	M1	YAML schema, validator, tolerance catalog
`c7_M2_golden_collector.py`	M2	∼309 golden values (JSON database)
`c7_M3_regression_runner.py`	M3	Pass/fail report, failure injection
`c7_M4_cross_pillar_checker.py`	M4	7 CP checks, deviation report
`c7_M5_community_portal.py`	M5	Submission validator, gallery
`c7_TC1_tolerance_benchmark.py`	TC1	L1+L2+L4; Figs. 9, 10
`c7_TC2_metrology_benchmark.py`	TC2	L1+L3+L4; Figs. 13, 14
`c7_TC3_system_benchmark.py`	TC3	L1+L3+L4; Figs. 17–19

14 Conclusion and Perspectives

14.1 Key Findings

C7 establishes the first cross-domain validation infrastructure for photonic circuit simulation spanning the classical-to-quantum spectrum. Four key findings:

1. **The four-level hierarchy is necessary and sufficient.** L1 (regression) catches code-level bugs; L2 (cross-tool) validates the PI speedup claim; L3 (cross-pillar) catches inter-document inconsistencies that no single-document test can detect; L4 (invariant) enforces physics correctness. All four levels fire on the test cases—removing any one would leave a validation gap.
2. **Cross-pillar consistency (L3) reveals non-trivial physics.** The $S = F$ identity holds for low-disorder devices but breaks down quantitatively for high-disorder cases (PhC, stadium billiard) where parity effects and chaos disrupt the wavefront–fidelity connection [8, 40]. This is not a failure of the benchmark but a validation of the physics: the benchmark correctly distinguishes regimes where $S = F$ is tight from regimes where it is not.
3. **The benchmark-as-contract format enables community adoption.** The YAML specification is machine-parseable, human-readable, and version-controlled. Any third party can implement their own test runner and verify conformance without access to the C-series source code.
4. **Binding-channel identification is reproducible.** TC3 correctly reproduces Ch 3 (Connectivity) as the binding channel for the 800G transceiver, consistent with C6's finding that thermal tuning power dominates the co-packaged power envelope [4, 37].

14.2 Perspective: Three Tiers of Validation Infrastructure

Table 11: Perspective: validation infrastructure tiers.

Tier	Approach	Description	Key Improvement
T1	C7 (current)	Synthetic golden references, four-level hierarchy, seven CP checks, three test cases. Python test runner with YAML specification format.	First cross-domain benchmark suite for photonic circuits. Catches regression, cross-tool, cross-pillar, and invariant failures.
T2	Foundry-calibrated	Golden references from foundry metrology data (wafer-level roughness statistics, lot-to-lot variation). Experimental validation against C4/C6 lab measurements. Multi-language test runners (MATLAB, Julia).	Process-specific yield prediction. Bridges the simulation-to-fabrication gap. Enables vendor qualification against foundry-specific benchmarks.
T3	Digital-twin integration	Real-time validation against digital twin fed by in-line metrology [19]. Adaptive golden references that update per wafer lot. Continuous CI/CD pipeline integrated with foundry MES.	Predictive, lot-specific, autonomous. Golden references are living documents that track process drift.

14.3 Condensed Reference

Table 12: C7 key results summary.

FoM	Definition	Target	Status
Regression pass rate	L1 benchmarks passing	$> 99\%$	**Pass**
Cross-tool agreement	Max PI-vs-MC deviation	$< 5\%$	**Pass**
$S = F$ precision	$\lvert S - F \rvert$ (wavefront-limited)	< 0.01	**Pass**
Unitarity	$\lvert \det(U) - 1 \rvert$	$< 10^{-12}$	**Pass**
CP checks	7 cross-pillar checks	All pass	**Pass**
Golden coverage	C-series FoMs with golden refs	100%	**309/309**
Community portal	Gallery operational	Functional	**Yes**

Appendix: Figures of Merit

Table 13: C7 figures of merit.

FoM	Definition	Target	Order
Regression pass rate	Fraction of L1 benchmarks passing	$> 99\%$	$\sim$309
Cross-tool agreement	Max PI-vs-MC deviation	$< 5\%$ rel.	8 cases
$S = F$ precision	$\lvert S - F \rvert$ across applicable devices	< 0.01	$\sim$15
Cross-pillar deviation	Max deviation in any CP check	$<$stated tol.	7 checks
Unitarity violation	$\lvert \det(U) - 1 \rvert$ for all matrices	$< 10^{-12}$	$\sim$50
Golden coverage	C-series FoMs with golden	100%	$\sim$309
Community benchmarks	External contributions	> 0 at v1	Target 10+

Established Practice vs. C7's Contribution

Table 14: Established practice vs. C7's contribution.

Aspect	**Established practice**	**C7's contribution**
Regression testing	Standard in software (pytest [11], CI/CD)	Applied to photonic sim. with physics-aware tolerance
Golden references	NIST SRD [23]; scattered photonic datasets [21]	Unified database: classical-to-quantum, 309 values
Cross-tool validation	Round-robin (EMPIRe [22])	Formalized as automated YAML-specified checks
Physics identity	Ad hoc in individual papers	Systematic L4: unitarity, $S = F$, resource hierarchy
Community submission	Rare in photonics; MLPerf [12], OpenML [13]	Structured format modeled on ML best practices

References

[1] J.-L. Chern, "From Fresnel to Fock Space: Worked Examples," *Engineering Optics Series* C1 v12, 2026.

[2] J.-L. Chern, "Path-Integral Tolerance Analysis in Optical Engineering," *Engineering Optics Series* C2 v7, 2026.

[3] J.-L. Chern, "Quantum Metrology & Verification in Optical Engineering," *Engineering Optics Series* C4 v1.2, 2026.

[4] J.-L. Chern, "System Integration for Photonic Circuits," *Engineering Optics Series* C6 v7, 2026.

[5] J.-L. Chern, "Production Toolkit Architecture & Shared Library," *Engineering Optics Series* C0 (scope), 2026.

[6] J.-L. Chern, "The Algorithmic Lens: Design Optimization," *Engineering Optics Series* B2 v7, 2026.

[7] J.-L. Chern, "Cooperativity & Entanglement," *Engineering Optics Series* B4 v8, 2026.

[8] J.-L. Chern, "Chaos & Complexity," *Engineering Optics Series* B5 v58, 2026.

[9] ISO/IEC 17025:2017, "General requirements for the competence of testing and calibration laboratories," 2017.

[10] JCGM 100:2008, "Evaluation of measurement data—Guide to the expression of uncertainty in measurement (GUM)," 2008.

[11] pytest development team, "pytest: helps you write better programs," https://pytest.org, 2023.

[12] P. Mattson *et al.*, "MLPerf Training Benchmark," *Proc. MLSys*, 2020.

[13] J. Vanschoren *et al.*, "OpenML: networked science in machine learning," *SIGKDD Explorations*, vol. 15, no. 2, pp. 49–60, 2013.

[14] Synopsys, "RSoft Photonic Device Tools," https://www.synopsys.com/photonic-solutions.html, 2023. (Now Keysight)

[15] Synopsys, "CODE V Optical Design Software," https://www.synopsys.com/optical-solutions/codev.html, 2023. (now keysight)

[16] Ansys, "Lumerical Photonic Simulation Software," https://www.lumerical.com, 2023.

[17] W. Bogaerts *et al.*, "Silicon microring resonators," *Laser Photon. Rev.*, vol. 6, no. 1, pp. 47–73, 2012.

[18] L. Chrostowski and M. Hochberg, *Silicon Photonics Design*, Cambridge Univ. Press, 2015.

[19] Z. Lu *et al.*, "Performance prediction for silicon photonics integrated circuits with layout-dependent correlated manufacturing variability," *Opt. Express*, vol. 25, pp. 9712–9733, 2017.

[20] P. P. Absil *et al.*, "Silicon photonics integrated circuits: a manufacturing platform for high density, low power optical I/O's," *Opt. Express*, vol. 23, pp. 9369–9378, 2015.

[21] JCMwave GmbH, "Benchmark suite for computational nano-optics," https://www.jcmwave.com/benchmarks, 2019.

[22] EMPIRe project, "European Metrology Programme for Innovation and Research: photonic circuit benchmarks," 2018.

[23] NIST Standard Reference Data Program, https://www.nist.gov/srd, 2024.

[24] C. K. Hong, Z. Y. Ou, and L. Mandel, "Measurement of subpicosecond time intervals between two photons by interference," *Phys. Rev. Lett.*, vol. 59, pp. 2044–2046, 1987.

[25] A. Politi *et al.*, "Silica-on-silicon waveguide quantum circuits," *Science*, vol. 320, pp. 646–649, 2008.

[26] J. Wang *et al.*, "Integrated photonic quantum technologies," *Nature Photon.*, vol. 14, pp. 273–284, 2020.

[27] M. Reck *et al.*, "Experimental realization of any discrete unitary operator," *Phys. Rev. Lett.*, vol. 73, pp. 58–61, 1994.

[28] W. R. Clements *et al.*, "Optimal design for universal multiport interferometers," *Optica*, vol. 3, pp. 1460–1465, 2016.

[29] V. Giovannetti, S. Lloyd, and L. Maccone, "Advances in quantum metrology," *Nature Photon.*, vol. 5, pp. 222–229, 2011.

[30] M. Born and E. Wolf, *Principles of Optics*, 7th ed., Cambridge Univ. Press, 1999.

[31] C. Sun *et al.*, "Single-chip microprocessor that communicates directly using light," *Nature*, vol. 528, pp. 534–538, 2015.

[32] M. Wade *et al.*, "TeraPHY: A chiplet technology for low-power, high-bandwidth in-package optical I/O," *IEEE Micro*, vol. 40, pp. 63–71, 2020.

[33] S. Shekhar *et al.*, "Roadmapping the next generation of silicon photonics," *Nature Commun.*, vol. 15, 751, 2024.

[34] P. Sibson *et al.*, "Chip-based quantum key distribution," *Nature Commun.*, vol. 8, 13984, 2017.

[35] T. K. Paraíso *et al.*, "A photonic integrated quantum secure direct communication system," *Nature Photon.*, vol. 15, pp. 850–856, 2021.

[36] G. Zhang *et al.*, "An integrated silicon photonic chip platform for continuous-variable quantum key distribution," *Nature Photon.*, vol. 17, pp. 979–986, 2023.

[37] M. Jacques *et al.*, "Optimization of thermo-optic phase shifter design and mitigation of thermal crosstalk," *Opt. Express*, vol. 27, pp. 10456–10471, 2019.

[38] N. C. Harris *et al.*, "Efficient, compact and low loss thermo-optic phase shifter in silicon," *Opt. Express*, vol. 22, pp. 10487–10493, 2014.

[39] P. Dong *et al.*, "Thermally tunable silicon racetrack resonators with ultralow tuning power," *Opt. Express*, vol. 18, pp. 20298–20304, 2010.

[40] T. A. Brody *et al.*, "Random-matrix physics: spectrum and strength fluctuations," *Rev. Mod. Phys.*, vol. 53, pp. 385–479, 1981.

[41] H.-S. Zhong *et al.*, "Quantum computational advantage using photons," *Science*, vol. 370, pp. 1460–1463, 2020.

[42] L. S. Madsen *et al.*, "Quantum computational advantage with a programmable photonic processor," *Nature*, vol. 606, pp. 75–81, 2022.

[43] C. Taballione *et al.*, "20-mode universal quantum photonic processor," *Quantum*, vol. 7, 1071, 2023.